

Digital Signal Processing Using Automotive Engine Signals as a Guiding Example

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1 The ECU Sees Only Signals

Modern engineering systems with controls rely on information obtained from measurements. Whether we consider a communication system, a radar, a medical device, or an automotive engine control unit, decisions are rarely based on direct observation of the underlying physical processes.

Instead, information is acquired through sensors and represented as electrical signals. The objective of signal processing is to extract meaningful information from these signals and to transform raw measurements into useful knowledge about the system under observation.

This lecture introduces the fundamental concepts and methods of digital signal processing using a modern spark-ignition engine as a guiding example. Throughout the course, we will investigate how an engine control unit estimates engine speed, crankshaft position, combustion quality, air-fuel ratio, and knock intensity solely from sensor data.

A central theme will accompany us throughout the entire lecture:

The ECU does not see the engine. It sees only signals.

Understanding how information is encoded in signals and how it can be recovered through mathematical and computational methods is the primary goal of digital signal processing.

1.1 Learning Objectives

After completing this chapter, the reader should be able to:

- explain why modern engine control units rely entirely on sensor signals to obtain information about the engine state,
- identify the main sensors used in a spark-ignition engine and describe the physical quantities they measure,
- define the term *signal* from an engineering perspective,
- distinguish between continuous-time and discrete-time signals,
- distinguish between deterministic and stochastic signals,
- distinguish between periodic and aperiodic signals,
- recognize that many practical engine control problems can be formulated as signal processing tasks,
- understand the role of digital signal processing in extracting information from measured sensor data.

1.2 The Challenge of Engine Control

Modern spark-ignition engines operate under highly dynamic conditions. At every instant, the engine control unit (ECU) must decide

- when to inject fuel,
- when to ignite the air-fuel mixture,
- whether combustion knock is present,
- and how to optimize efficiency while complying with emission limits.

At first glance, these decisions appear to require direct knowledge of the combustion process occurring inside the cylinders. However, the ECU has no direct access to the physical processes taking place within the engine.

The ECU cannot see the pistons. It cannot observe the flame front. It cannot directly measure the pressure distribution inside the combustion chamber.

Instead, all available information reaches the ECU through electrical sensor signals.

Consequently, every engine control decision ultimately depends on the acquisition, processing, and interpretation of signals.

*The ECU does not see the engine.
It sees only signals.*

1.3 Available Sensor Signals

The engine control unit (ECU) must continuously estimate the operating state of the engine in order to perform fuel injection, ignition timing control, knock detection, and emission control.

However, the ECU has no direct access to the physical processes occurring inside the engine. Instead, all available information is provided by sensors that convert physical quantities into electrical signals.

Figure 1.1 illustrates this basic concept. Each sensor provides a different view of the en-



Figure 1.1: Information flow from the physical engine to the ECU.

gine state. Together, these signals form the basis for all control and diagnostic functions.

1.3.1 Crankshaft Position Sensor

The crankshaft position sensor measures the rotational movement and the crank angle of the crankshaft.

In modern engines, a toothed wheel with a missing-tooth pattern (e.g. a 60-2 wheel) is commonly used. As the teeth pass a Hall-effect or inductive sensor, an electrical pulse train is generated. From this signal, the ECU can determine

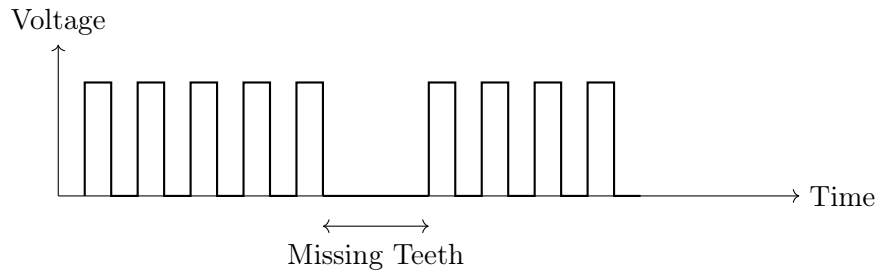


Figure 1.2: Idealized crankshaft position signal generated by a 60-2 wheel. The enlarged gap identifies a unique crankshaft position and serves as a synchronization marker for the ECU.

- crankshaft position,
- rotational speed,
- angular acceleration.

The crankshaft signal is one of the most important signals in the engine management system because many control functions are synchronized to the crankshaft angle.

1.3.2 Camshaft Position Sensor

The camshaft position sensor provides phase information that allows the ECU to distinguish between the different cylinders and to determine valve opening times.

While the crankshaft sensor indicates angular position, the camshaft sensor identifies the current engine cycle. The crankshaft rotates 720 degrees in one cycle and the camshaft rotates 360 degrees in one cycle.

Together, crankshaft and camshaft signals enable precise timing of fuel injection and ignition events.

1.3.3 Knock Sensor

Combustion knock is an abnormal combustion phenomenon characterized by high-frequency pressure oscillations inside the combustion chamber.

A piezoelectric knock sensor mounted on the engine block converts these vibrations into an electrical signal.

The resulting signal contains information about

- combustion-induced vibrations,
- engine mechanical noise,
- potential knocking events.

Since knock-related components are often hidden within other vibration sources, signal processing techniques are required to extract the relevant information.

1.3.4 Ion Current Signal

Under suitable operating conditions, the spark plug can be used not only as an ignition device but also as a sensor. After ignition, charged particles generated during combustion allow a small electrical current to flow through the combustion chamber. This ion current signal contains information about

- ignition timing,
- flame propagation,
- combustion stability,
- combustion phasing.

Unlike the knock sensor, which observes engine vibrations indirectly, the ion current signal provides a direct view of the combustion process inside the cylinder.

1.3.5 Lambda Sensor

The lambda sensor measures the residual oxygen concentration in the exhaust gas. From this measurement, the ECU can estimate whether the air-fuel mixture is rich or lean relative to the stoichiometric operating point. The lambda signal is essential for

- fuel mixture control,
- emission reduction,
- catalytic converter efficiency.

Compared to the crankshaft and knock sensor signals, the lambda signal typically changes more slowly and is primarily used for closed-loop control.

Physical Meaning

The lambda value is defined as

$$\lambda = \frac{\text{actual air-fuel ratio}}{\text{stoichiometric air-fuel ratio}}. \quad (1.1)$$

The interpretation is straightforward:

- $\lambda < 1$ indicates a rich mixture,
- $\lambda = 1$ corresponds to a stoichiometric mixture,
- $\lambda > 1$ indicates a lean mixture.

The lambda sensor converts this information into an electrical signal that can be processed by the ECU.

Typical Sensor Signal

Unlike the crankshaft or knock sensor signals, the lambda sensor typically varies relatively slowly. For a conventional switching-type lambda sensor, the output voltage changes rapidly when the mixture crosses the stoichiometric operating point. A typical signal may look as follows: The ECU continuously monitors this signal and adjusts the fuel

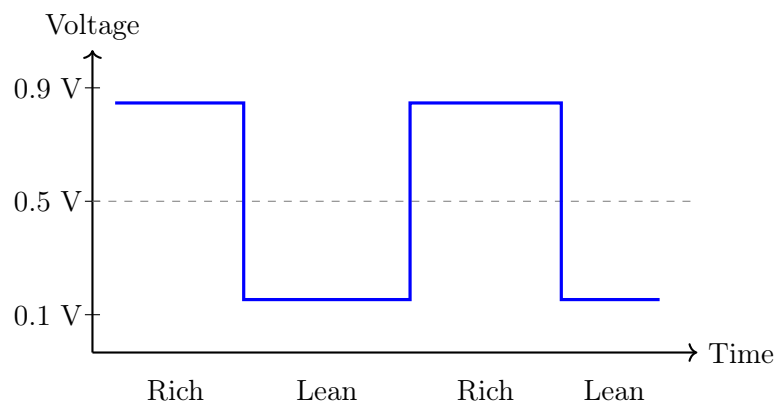


Figure 1.3: Typical output signal of a switching-type lambda sensor. The sensor voltage changes rapidly when the air-fuel ratio crosses the stoichiometric operating point.

injection quantity in order to keep the engine operating near $\lambda = 1$.

A Different Type of Signal

The lambda sensor illustrates an important observation: Not all signals encountered in engine control have the same characteristics. The crankshaft signal is highly periodic and directly related to engine rotation. The knock sensor signal contains broadband vibration components and requires frequency-domain analysis. The ion current signal provides information about individual combustion events. The lambda signal, in contrast, changes comparatively slowly and is primarily used for feedback control rather than frequency analysis. Consequently, different signal processing methods are required for different types of sensor signals.

Why This Signal Matters

The lambda sensor is one of the most important sensors in modern engine management systems because it closes the control loop between combustion and fuel injection. Without this signal, the ECU would have no direct information about whether the supplied fuel quantity results in the desired air-fuel ratio. As with all other examples discussed so far, the ECU does not directly observe the physical process itself. Instead, it must infer the state of the engine from an electrical signal provided by a sensor. The lambda sensor therefore provides another example of the central theme of this lecture:

1.3.6 Summary of Available Signals

The sensors discussed above provide different types of information and exhibit very different signal characteristics. Although these sensors measure different physical quantities,

Sensor	Measured Quantity	Primary Information
Crankshaft Sensor	Rotational position	Engine speed and angle
Camshaft Sensor	Camshaft position	Cylinder identification
Knock Sensor	Engine vibration	Combustion knock
Ion Current Sensor	Combustion ionization	Combustion process
Lambda Sensor	Residual oxygen	Air-fuel ratio

Table 1.1: Typical sensor signals available to an engine control unit.

they all have one thing in common: each sensor provides information in the form of an electrical signal. The extraction of useful information from these signals is the central task of digital signal processing.

Modern spark-ignition engines operate most efficiently and produce the lowest emissions when the air-fuel mixture is close to its stoichiometric value. To determine whether the mixture is too rich or too lean, the ECU evaluates the signal generated by the lambda sensor.

The ECU does not see the engine. It sees only signals.

1.4 What is a Signal?

After examining several engine sensors, a natural question arises:

What exactly is a signal?

Although the sensor signals discussed so far originate from very different physical phenomena, they share a common purpose: they convey information about the state of the engine.

In engineering, a signal is therefore understood as a quantity that varies with one or more independent variables and carries information about a physical process.

1.4.1 Definition of a Signal

A signal can generally be defined as

$$x = x(\xi) \quad (1.2)$$

where ξ denotes an independent variable. In many engineering applications, the independent variable is time, leading to the familiar representation

$$x(t). \quad (1.3)$$

The value of the signal at each instant of time contains information about the underlying physical system. For example, the voltage generated by a crankshaft sensor changes as the crankshaft rotates, while the voltage generated by a knock sensor changes according to the vibration state of the engine block.

1.4.2 Examples from Engine Control

The sensors introduced previously produce different signals that can be represented mathematically. The crankshaft sensor produces a signal

$$x_c(t), \quad (1.4)$$

which contains information about crankshaft position and engine speed. The knock sensor produces a signal

$$x_k(t), \quad (1.5)$$

which reflects vibration and acoustic phenomena within the engine. The ion current signal may be represented as

$$x_i(t), \quad (1.6)$$

providing information about combustion events inside the cylinder. Similarly, the lambda sensor generates a signal

$$x_\lambda(t), \quad (1.7)$$

which is related to the air-fuel ratio of the engine. Although these signals originate from different sensors and represent different physical quantities, they can all be treated using the same mathematical framework. This observation forms the foundation of signal processing.

1.4.3 Signals as Carriers of Information

The numerical value of a signal is usually not the primary quantity of interest. Instead, the information encoded within the signal is important. For example, the ECU is generally not interested in the voltage level generated by the crankshaft sensor itself. Rather, it seeks to extract information such as

- crankshaft position,

- rotational speed,
- combustion timing,
- knock intensity,
- air-fuel ratio.

Signal processing can therefore be viewed as the process of converting raw measurements into useful information.

Signals are not the final objective. They are the medium through which information about a physical system is conveyed.

This idea will accompany us throughout the remainder of this lecture. Nearly every signal processing technique introduced later can be understood as a method for extracting information from measured signals.

1.5 Signal Classification

The sensor signals encountered in an engine management system exhibit very different characteristics. Some signals are highly periodic, others appear random. Some are measured continuously, while others are available only at discrete points in time. Classifying signals according to their properties helps us choose appropriate analysis and processing methods. In the following sections, several important signal classes are introduced using examples from engine control systems.

1.5.1 Continuous-Time and Discrete-Time Signals

A continuous-time signal is defined for every instant of time. It is commonly represented as

$$x(t). \quad (1.8)$$

Examples include the analog voltage generated by a knock sensor or the analog output of a lambda sensor. Figure ?? illustrates a typical continuous-time signal. A discrete-time signal is defined only at specific sampling instants

$$x[n]. \quad (1.9)$$

The integer index n denotes the sample number rather than physical time. Discrete-time signals arise naturally when analog sensor signals are converted into digital form by an analog-to-digital converter (ADC). For example, the analog knock sensor signal

$$x_k(t) \quad (1.10)$$

may be converted into the discrete sequence

$$x_k[n]. \quad (1.11)$$

Most signal processing algorithms implemented in an ECU operate on discrete-time signals.

1.5.2 Deterministic and Random Signals

A deterministic signal can be predicted exactly if its mathematical description is known. For example, an ideal sinusoidal signal

$$x(t) = A \sin(\omega t) \quad (1.12)$$

is deterministic because its future values are completely determined by the parameters A and ω . The crankshaft position signal of an engine operating at constant speed is approximately deterministic and highly predictable. In contrast, a random (or stochastic) signal contains unpredictable components. The knock sensor signal provides a typical example. Although certain frequency components may be associated with knocking combustion, the measured signal also contains

- mechanical vibrations,
- sensor noise,
- combustion variability,
- environmental disturbances.

As a result, the exact waveform cannot be predicted precisely. Many practical DSP methods are specifically designed to extract useful information from random signals.

1.5.3 Periodic and Aperiodic Signals

A signal is periodic if it repeats after a fixed period T :

$$x(t) = x(t + T). \quad (1.13)$$

The crankshaft position signal generated by a rotating 60-2 wheel is a typical example of a nearly periodic signal. As long as the engine speed remains approximately constant, the signal pattern repeats every revolution. Periodic signals play a central role in signal processing because they can often be analyzed efficiently in the frequency domain. An aperiodic signal does not repeat exactly. Examples include

- a single combustion event,
- a transient knock occurrence,
- a sudden throttle change,
- engine start-up.

The ion current signal associated with a single combustion cycle is a typical aperiodic signal.

1.5.4 Analog and Digital Signals

An analog signal can assume any value within a continuous range. For example, the voltage produced by a knock sensor varies continuously. A digital signal, on the other hand, can assume only a finite number of values. Digital signals are obtained by sampling and quantization of analog measurements. Since modern ECUs are digital systems, nearly all signal processing algorithms operate on digital representations of sensor signals. The conversion from analog to digital signals will be discussed in detail in later chapters.

1.5.5 Classification of Engine Signals

The engine signals introduced earlier can now be classified according to the concepts developed in this section. The diversity of these signal types illustrates an important fact:

Signal	Time Domain	Predictability	Periodicity
Crankshaft Sensor	Continuous / Discrete	Mostly Deterministic	Periodic
Camshaft Sensor	Continuous / Discrete	Deterministic	Periodic
Knock Sensor	Continuous / Discrete	Stochastic	Quasi-periodic
Ion Current Signal	Continuous / Discrete	Partly Deterministic	Aperiodic
Lambda Sensor	Continuous / Discrete	Slowly Varying	Aperiodic

Table 1.2: Classification of typical engine-related signals.

There is no universal signal processing method that is optimal for all signals. Instead, the choice of analysis and processing techniques depends strongly on the characteristics of the signal under investigation.

1.6 Why Signal Processing?

The sensor signals discussed throughout this chapter contain valuable information about the operating state of the engine. However, this information is rarely available in a directly usable form. The ECU does not receive quantities such as

- engine speed,
- crankshaft position,
- combustion quality,
- knock intensity,
- air-fuel ratio

as explicit numerical values. Instead, it receives electrical sensor signals from which these quantities must be estimated. Signal processing provides the mathematical and computational tools required to perform this estimation. For example,

- crankshaft speed is obtained from the timing of crankshaft sensor pulses,
- knock intensity is estimated from vibration measurements of the knock sensor,
- combustion characteristics can be extracted from ion current signals,
- air-fuel ratio information is obtained from the lambda sensor.

In each case, the objective is the same:

Raw Signal $\rightarrow$ Information

Digital signal processing (DSP) develops systematic methods for performing this transformation. The remaining chapters of this lecture introduce the mathematical concepts required for this task. Beginning with simple signal and system models, we will gradually develop the tools needed to analyze, transform, filter, and interpret engine-related signals.

The ECU does not see engine speed. It does not see combustion. It does not see knock. It sees only signals and must infer everything else.

1.7 Summary

Modern engine control systems rely entirely on information obtained from sensors. The ECU has no direct access to the physical processes occurring inside the engine and therefore must estimate the engine state from measured signals. Several important sensor signals were introduced:

- crankshaft position sensor,
- camshaft position sensor,
- knock sensor,
- ion current signal,
- lambda sensor.

A signal was defined as a quantity that varies with an independent variable and carries information about a physical process. Although engine sensors measure different physical quantities, their outputs can be represented using a common mathematical framework. The chapter also introduced several important signal classifications, including

- continuous-time and discrete-time signals,
- deterministic and stochastic signals,
- periodic and aperiodic signals,
- analog and digital signals.

These concepts provide the foundation for all subsequent chapters. The central message of this chapter can be summarized in a single sentence:

The ECU does not see the engine. It sees only signals.

The objective of digital signal processing is to extract useful information from those signals.

1.8 Exercises

1. Explain why an ECU cannot directly observe the combustion process inside a cylinder.
2. Identify the primary purpose of each of the following sensors:
 - a) crankshaft position sensor,
 - b) camshaft position sensor,
 - c) knock sensor,
 - d) lambda sensor.
3. Give three examples of information that must be estimated from sensor signals in a modern engine management system.
4. Explain the difference between a continuous-time signal and a discrete-time signal.
5. Explain the difference between a deterministic signal and a stochastic signal.
6. Is the crankshaft position signal periodic or aperiodic? Justify your answer.
7. Why is the knock sensor signal generally more difficult to interpret than the crankshaft sensor signal?
8. The ECU receives a sequence of sampled values from an analog sensor. Is this signal
 - a) continuous-time or discrete-time?
 - b) analog or digital?

Explain your reasoning.

9. Consider the statement:

The ECU does not see the engine. It sees only signals.

Discuss the meaning of this statement in the context of engine control.

10. Give an example from everyday life of a system that relies on signals rather than direct observation.

2 Systems

In the previous chapter, signals were introduced as carriers of information about physical processes. However, signals rarely exist in isolation. In engineering applications, signals are generated, transformed, transmitted, filtered, and interpreted by systems. A modern engine control unit can therefore be viewed as a collection of interconnected systems that process sensor signals and generate control actions. To understand signal processing, we must first understand the concept of a system.

2.1 Why Study Systems?

The ECU receives signals from sensors and uses them to make control decisions. Between measurement and decision, the signals pass through a sequence of processing steps. For example:

- the crankshaft signal is processed to determine engine speed,
- the knock sensor signal is analyzed to detect combustion knock,
- the lambda signal is evaluated to adjust fuel injection.

Each processing step can be regarded as a system that transforms an input signal into an output signal. Consequently, many engineering problems can be formulated as

$$\text{Input Signal} \rightarrow \text{System} \rightarrow \text{Output Signal}$$

This viewpoint forms the foundation of modern signal processing.

2.2 What Is a System?

A system is an entity that transforms one signal into another. If the input signal is denoted by

$$x(t)$$

and the output signal by

$$y(t),$$

then the action of the system may be written as

$$y(t) = \mathcal{S}\{x(t)\},$$

where

$$\mathcal{S}$$

denotes the system operator.

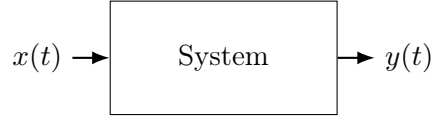


Figure 2.1: A system transforms an input signal into an output signal.

2.2.1 Throttle Position and Engine Torque

The driver changes the throttle position. The engine responds by producing torque. The engine can therefore be viewed as a system:

$$\text{Throttle Position} \rightarrow \text{Engine} \rightarrow \text{Torque}.$$

2.2.2 Fuel Injection and Air-Fuel Ratio

A change in injected fuel mass affects the combustion process and therefore the measured lambda value. This relationship can be represented as

$$\text{Fuel Injection} \rightarrow \text{Engine} \rightarrow \lambda.$$

2.2.3 Combustion and Knock Sensor Signal

Pressure oscillations generated during combustion propagate through the engine structure. The engine block and the knock sensor transform these oscillations into an electrical signal. Hence,

$$\text{Combustion} \rightarrow \text{Engine Block} \rightarrow \text{Knock Sensor Signal}.$$

2.3 Mathematical Description of Systems

So far, systems have been introduced qualitatively as entities that transform input signals into output signals. For engineering analysis, however, a more precise mathematical description is required. In general, a system establishes a relationship between an input signal and an output signal. If the input signal is denoted by

$$x(t)$$

and the output signal by

$$y(t),$$

then a system can be represented symbolically as

$$y(t) = \mathcal{S}x(t),$$

where ($\mathcal{S}$) denotes the system operator. The operator specifies how the input signal is transformed into the output signal. For example, consider the simple system

$$y(t) = 2x(t).$$

This system multiplies every input value by two. If the input is

$$x(t) = 3,$$

the output becomes

$$y(t) = 6.$$

Although this example is simple, it illustrates an important idea: A system is defined by the rule that maps inputs to outputs. In practice, engineering systems are often considerably more complex. For example, the relationship between injected fuel mass and the resulting lambda value involves combustion dynamics, gas exchange, transport delays, and sensor characteristics. Nevertheless, the same input-output viewpoint remains valid. Throughout this lecture, systems will be studied primarily through their effect on signals. The internal physical mechanisms are often less important than the relationship between the observable input and output quantities.

2.4 Static and Dynamic Systems

Not all systems respond to their inputs in the same way. One of the most important distinctions is the difference between static and dynamic systems. A static system, sometimes called a memoryless system, produces an output that depends only on the current value of the input. For example, $y(t)=2x(t)$ is a static system. At every instant, the output is determined solely by the current input value. No information about past inputs is required. In contrast, a dynamic system possesses memory. Its output depends not only on the current input but also on previous input values. Many physical systems are dynamic because they store energy, mass, or momentum. A simple example from engine control is the rotational motion of the crankshaft. Suppose the injected fuel quantity is suddenly increased. The engine speed does not increase instantaneously. Instead, the rotational inertia of the engine causes the speed to change gradually. The current engine speed therefore depends on both the present fuel input and the previous operating history. Consequently, the engine behaves as a dynamic system. Most systems encountered in signal processing are dynamic. Signals are often filtered, delayed, integrated, differentiated, or otherwise modified in ways that depend on their past values. Understanding dynamic behavior is therefore one of the central goals of system theory.

2.5 Causal and Non-Causal Systems

Another important classification concerns causality. A system is said to be causal if its output at a given time depends only on present and past input values. In other words, a causal system cannot use information from the future. This property can be expressed conceptually as

$$y(t) = f(x(\tau), \tau \leq t).$$

All physical systems operating in real time are causal. An engine control unit provides an obvious example. When determining the ignition timing for the next combustion

cycle, the ECU can only use sensor measurements that have already been acquired. Future measurements are not yet available. Consequently, all real-time control algorithms implemented in an ECU must be causal. In contrast, a non-causal system depends on future input values. Such systems cannot be implemented in real time but may be useful for offline signal analysis. For example, when analyzing previously recorded knock sensor data on a computer, it is possible to process the signal using information from both the past and the future. Although non-causal systems are important in certain applications, causal systems are of primary interest in real-time digital signal processing.

2.6 Memoryless Systems and Systems with Memory

The concept of memory is closely related to the distinction between static and dynamic systems. A memoryless system produces an output that depends only on the input value at the same instant. Mathematically,

$$y(t) = f(x(t)).$$

Examples include simple amplifiers and scaling operations. If the current input value is known, the output can be computed immediately. No information about previous inputs is required. A system with memory, on the other hand, depends on input values from earlier times. For example, consider a moving-average operation:

$$y[n] = \frac{1}{3} (x[n] + x[n-1] + x[n-2]).$$

The output depends on the current sample and on two previous samples. The system therefore possesses memory. Many systems encountered in engine control exhibit this behavior. For example, engine speed estimation is often performed by averaging multiple crankshaft measurements in order to reduce noise. Similarly, lambda control algorithms frequently use previous sensor measurements to improve stability and robustness. Systems with memory are fundamental to signal processing because they enable filtering, smoothing, prediction, and many other useful operations.

2.7 Looking Ahead

The systems discussed in this chapter have been introduced in a general manner. In principle, a system can possess arbitrary properties and may be described by highly complex mathematical relationships. Fortunately, many practical systems encountered in signal processing belong to a particularly important class known as linear time-invariant systems. These systems satisfy additional mathematical properties that greatly simplify analysis and design. They form the theoretical foundation of much of modern signal processing and are especially important for filtering, frequency analysis, and control applications. The next chapter therefore focuses on linear time-invariant systems and develops the mathematical tools required to analyze them.